# AUSP1FY INTERNSHIP — TASK 6

## Netflix Content Analytics & Business Insights Dashboard

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**Role:** Data Analyst / Data Science Intern
**Project Type:** Data Analysis, Machine Learning and Business Intelligence
**Technology:** Python, Pandas, Matplotlib, Scikit-learn, Streamlit

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# 1. Introduction

This project focuses on analyzing a Netflix content dataset to identify meaningful patterns, trends and business insights.
The project combines data cleaning, exploratory data analysis, data visualization, machine learning and interactive dashboard development into a single data analytics solution.
An interactive Streamlit dashboard was developed to allow users to explore the Netflix catalogue, filter records, examine trends, analyze business insights and use a machine-learning model to predict whether a title is a Movie or TV Show.

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# 2. Project Objectives

The main objectives of this project were to:

Analyze the structure and characteristics of the Netflix dataset.

Identify and handle missing values and duplicate records.
Examine the distribution of Movies and TV Shows.

Analyze content ratings, countries and genres.

Investigate content release trends over time.

Analyze movie durations and TV Show seasons.

Develop a machine-learning classification model.

Evaluate the performance of the machine-learning model.

Create an interactive business analytics dashboard.

Generate data-driven business insights and recommendations.

Provide an executive report that can be downloaded from the dashboard.

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# 3. Dataset Overview

The dataset contains information about Netflix titles, including:

Title

Content type

Director

Cast

Country

Date added

Release year

Rating

Duration

Listed genres

Description

The dataset contains approximately **8,790 records**.
The two main content categories are:

| Content Type | Number of Titles | Percentage |
|---|---|---|
| Movie | 6,126 | 69.69% |
| TV Show | 2,664 | 30.31% |
| Total | 8,790 | 100% |

Movies therefore represent the larger portion of the dataset, while TV Shows represent approximately one-third of the catalogue.

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# 4. Tools and Technologies

The project was developed using the following technologies:

## Python

Used as the main programming language for data processing, analysis and machine learning.

## Pandas

Used for:

Loading the dataset

Data cleaning

Missing-value analysis

Filtering

Grouping

Sorting

Statistical analysis

Data transformation

## Matplotlib

Used to create analytical visualizations and the machine-learning confusion matrix.

## Scikit-learn

Used for:

Train/test splitting

Data preprocessing

One-hot encoding

Pipeline creation

Random Forest classification

Model evaluation

### Joblib
Used to save and load the trained machine-learning model.
### Streamlit
Used to develop the interactive web-based analytics dashboard.
### VS Code
Used as the main development environment.

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# 5. Data Cleaning and Preparation
The dataset was first inspected to understand its structure, columns and data types.
The cleaning process included:

Checking dataset dimensions.

Inspecting column names.

Checking data types.

Identifying missing values.

Detecting duplicate records.

Removing duplicate records.

Filling missing text values where appropriate.

Filling numerical missing values using median values.

Preparing the cleaned dataset for analysis and machine learning.

Text fields such as director, genre information and country were handled appropriately so that missing values would not prevent further analysis.
The cleaned dataset was then used for exploratory analysis and machine-learning preparation.

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# 6. Exploratory Data Analysis
## 6.1 Content Type Distribution
The dataset contains:

6,126 Movies

2,664 TV Shows

Movies account for **69.69%** of the dataset, while TV Shows account for **30.31%**. This shows that the dataset contains substantially more Movies than TV Shows.

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## 6.2 Content Ratings

The most common ratings identified were:

| Rating | Number of Titles |
|---|---|
| TV-MA | 3,205 |
| TV-14 | 2,157 |
| TV-PG | 861 |
| R | 799 |
| PG-13 | 490 |

TV-MA was the most frequently occurring rating in the dataset.

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## 6.3 Top Countries

The countries with the highest number of titles included:

| Country | Titles |
|---|---|
| United States | 3,240 |
| India | 1,057 |
| United Kingdom | 638 |
| Pakistan | 421 |
| Canada | 271 |
| Japan | 259 |
| South Korea | 214 |
| France | 213 |
| Spain | 182 |

The United States had the highest number of titles in the dataset.

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## 6.4 Genre Analysis

The most common genres included:

| Genre | Titles |
|---|---|
| International Movies | 2,752 |
| Dramas | 2,426 |
| Comedies | 1,674 |
| International TV Shows | 1,349 |
| Documentaries | 869 |
| Action & Adventure | 859 |
| TV Dramas | 762 |
| Independent Movies | 756 |

| Genre | Titles |
|---|---|
| Children & Family Movies | 641 |
| Romantic Movies | 616 |

International Movies were the most frequently occurring genre category.

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## 6.5 Release Year Analysis

The dataset was also analyzed according to release year.
Some of the years with high numbers of titles included:

2017: 1,030 titles

2018: 1,146 titles

2019: 1,030 titles

2020: 953 titles

2021: 592 titles

The analysis showed that the dataset contains a substantial concentration of relatively recent content.

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## 6.6 Movie Duration

Movie duration was extracted from the duration field and converted into numerical values.
The results were:

Average movie duration: **99.58 minutes**

Median movie duration: **98 minutes**

Minimum duration: **3 minutes**

Maximum duration: **312 minutes**

The average movie duration provides a useful overview of the typical length of movies represented in the dataset.

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## 6.7 TV Show Seasons

TV Show duration was analyzed based on the number of seasons.
The results included:

Average: approximately **1.75 seasons**

Median: **1 season**

Minimum: **1 season**

Maximum: **17 seasons**

The median indicates that a typical TV Show in the dataset has one season.

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# 7. Machine Learning

A machine-learning classification system was developed to predict whether a Netflix title is a **Movie** or a **TV Show**.

## Target Variable

The target variable was:
**Content Type**
with two possible classes:

Movie

TV Show

## Features

The model used metadata including:

Release year

Rating

Country

Genre information

Categorical variables were transformed using **OneHotEncoder**.
A preprocessing and classification pipeline was created to combine the data transformation process with the machine-learning model.

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# 8. Machine Learning Model

The classification algorithm used was:
**Random Forest Classifier**
The dataset was divided into training and testing sets using an 80/20 split.

The model was trained using the training dataset and evaluated using the test dataset.
A saved model file was created:
`content_type_model.pkl`
This trained model is loaded by the Streamlit dashboard and used by the interactive prediction system.

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# 9. Model Evaluation

The Random Forest model produced the following results:

| Metric | Result |
|---|---|
| Accuracy | 96.99% |
| Precision | 96.66% |
| Recall | 99.10% |
| F1 Score | 97.86% |

These metrics were calculated using the held-out test dataset.
The confusion matrix was:

| | Predicted Movie | Predicted TV Show |
|---|---|---|
| Actual Movie | 1,214 | 11 |
| Actual TV Show | 42 | 491 |

The evaluation results are displayed directly within the interactive dashboard.

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# 10. Interactive Prediction System

The dashboard contains an interactive prediction tool.
Users can enter:

Release year

Rating

Country

Genres

The trained Random Forest model then predicts whether the provided metadata represents a:
**Movie** or **TV Show**
The dashboard also displays the model's prediction confidence and probability distribution for the two content classes.
This demonstrates how a trained machine-learning model can be integrated into a practical data application.

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# 11. Business Insights

The analysis generated several important findings.

### Content Distribution
Movies represent **69.69%** of the dataset, while TV Shows represent **30.31%**.
### Geographic Distribution
The United States has the highest number of titles in the dataset.
### Content Rating
TV-MA is the most common content rating.
### Genre
International Movies are the most common genre category.
### Movie Duration
The average movie duration is approximately **99.58 minutes**.
### TV Show Structure
The median TV Show has **1 season**.
These findings provide a high-level view of the composition and characteristics of the Netflix catalogue represented in the dataset.

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# 12. Business Recommendations
Based on the analysis, the following recommendations were generated:

Maintain strong movie content availability while continuing to develop the TV Show catalogue.

Continue monitoring international content because international categories represent a significant part of the dataset.

Consider audience segmentation when analyzing mature-rated content because TV-MA is the most common rating.

Use historical content-addition trends to support future content acquisition and release planning.

Consider both short-form and longer-running TV series when analyzing opportunities within the TV Show catalogue.

These recommendations are based on patterns identified in the analyzed dataset.

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# 13. Interactive Dashboard
A Streamlit dashboard was developed to make the analysis accessible through an interactive interface.
The dashboard contains four major sections.
## Overview
Provides:

Total titles

Movie percentage

TV Show percentage

Top rating

Content distribution

Rating analysis

Country analysis

Genre analysis

Release-year trends

Movie duration analysis

## Dataset Explorer

Allows users to:

Filter by country

Filter by rating

Search titles

View matching records

Download filtered data as CSV

## Machine Learning

Provides:

Model performance metrics

Confusion matrix

Model information

Interactive content prediction

Prediction confidence

Prediction probabilities

### Business Insights

Provides:

Business KPIs

Data-driven findings

Content characteristics

Business recommendations

Executive summary

Downloadable executive report

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## 14. Dashboard Filtering

The dashboard includes interactive controls that allow users to select:

Release-year range

Content type

The selected filters dynamically update the analysis.
This allows users to investigate different periods and content categories without modifying the underlying Python code.

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## 15. Executive Reporting

A downloadable executive report was integrated into the dashboard.
The report summarizes:

Analysis period

Content filter

Dataset size

Movie and TV Show distribution

Top country

Top rating

Top genre

Average movie duration

Median TV Show seasons

Machine-learning performance

Business recommendations

Executive summary

This makes the dashboard useful not only for exploration but also for communicating analytical findings.

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# 16. Project Architecture

The project was organized into separate components for maintainability.
The project includes modules for:

Data analysis

Machine learning

Business insights

Business recommendations

Visualization

Streamlit dashboard

Dataset storage

Model storage

Output reports

The machine-learning model was saved separately and loaded into the dashboard when required.
This modular structure makes the project easier to maintain and extend.

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# 17. Skills Demonstrated

This project demonstrates practical experience in:

Python programming

Pandas

Data cleaning

Exploratory Data Analysis

Data visualization

Statistical analysis

Feature preparation

Categorical encoding

Machine learning

Random Forest classification

Model evaluation

Streamlit dashboard development

Interactive data filtering

Business intelligence

Data storytelling

Business recommendations

Report generation

Modular Python development

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# 18. Challenges and Solutions

During development, several technical challenges were encountered.

### Data Cleaning

Missing values and duplicate records required systematic handling before analysis.

### Function Management

The project was separated into reusable functions so that data-loading, cleaning, analysis and machine-learning processes could be reused.

### Machine Learning

Categorical variables such as country, rating and genre required preprocessing before being supplied to the Random Forest model.

### Dashboard Integration

The trained model had to be loaded into Streamlit and connected to an interactive prediction interface.

### Data Filtering

The dashboard required filtering logic that could update KPIs and visualizations based on user selections.

These challenges helped strengthen the practical understanding of data-analysis workflows and application development.

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# 19. Future Improvements

Potential future improvements include:

Adding more machine-learning algorithms for comparison.

Adding advanced recommendation functionality.

Adding more interactive visualizations.

Adding automated data refresh.

Adding user authentication.

Adding database integration.

Adding more detailed time-series analysis.

Deploying the dashboard online.

Adding automated PDF report generation.

Adding advanced predictive analytics.

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# 20. Conclusion

This project demonstrates a complete data analytics workflow using a Netflix content dataset.
The project progressed from data inspection and cleaning to exploratory analysis, visualization, machine learning and business intelligence.
A Random Forest classification model achieved **96.99% accuracy**, with a precision of **96.66%**, recall of **99.10%** and F1 score of **97.86%** on the test dataset.
The final Streamlit application brings the entire workflow together in one interactive dashboard. Users can explore the dataset, analyze content trends, filter records, download data, evaluate the machine-learning model, make predictions and review business insights.
Overall, the project demonstrates how Python-based data analysis and machine learning can be transformed into an interactive business analytics solution.